

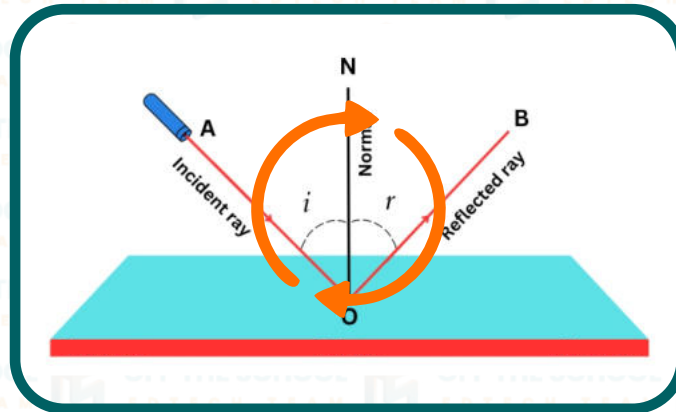
UNIT 13

GEOMETRICAL OPTICS



REFLECTION OF LIGHT

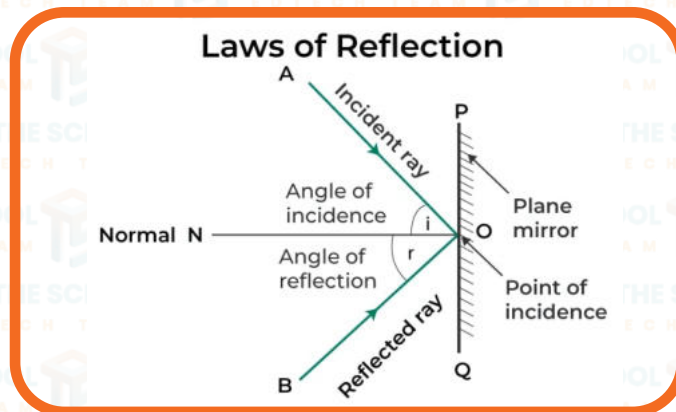
Reflection of light is the bouncing back of light rays when they encounter a surface, obeying the laws of reflection and resulting in the formation of images.



LAWS OF REFLECTION

FIRST LAW OF REFLECTION

The first law of reflection states that the incident ray, the reflected ray, and the normal (perpendicular line) to the reflecting surface all lie in the same plane.



SECOND LAW OF REFLECTION

The second law of reflection states that the angle of incidence is equal to the angle of reflection, measured between the incident ray and the normal compared to the reflected ray and the normal.

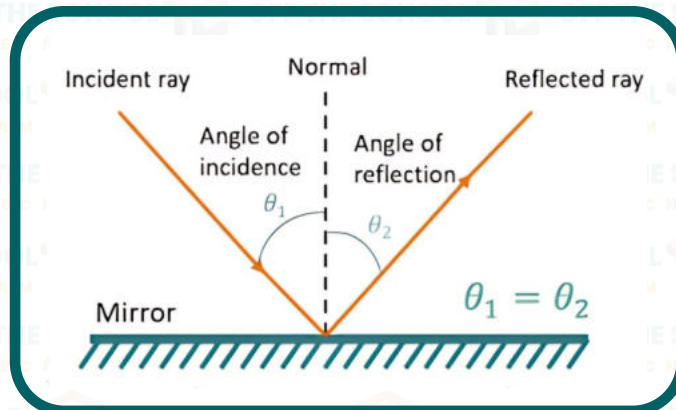


IMAGE LOCATION BY SPHERICAL MIRROR EQUATION

IMAGE FORMATION BY SPHERICAL MIRRORS

Spherical mirrors, whether concave or convex, create images through the reflection of light rays. The process varies based on the type of mirror:

CONCAVE MIRRORS

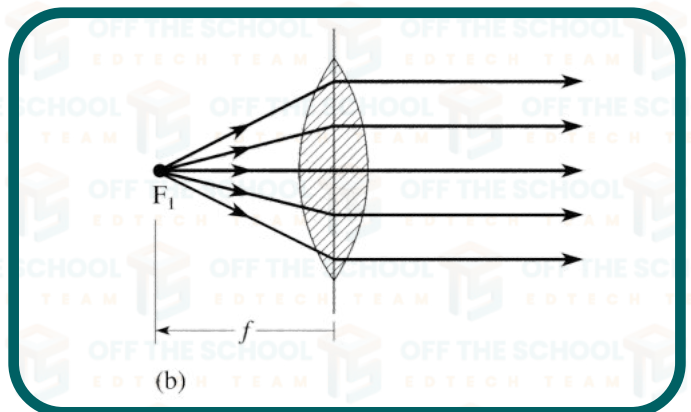
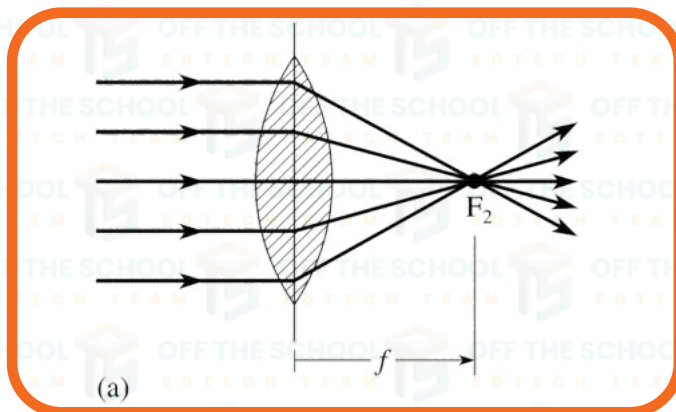
1. For objects beyond the focal point (F), real and inverted images are formed between the focal point and the mirror.
2. For objects at the center of curvature (C), real and inverted images are formed at the center of curvature.
3. For objects between the focal point and the mirror, real and inverted images are formed beyond the center of curvature.
4. For objects at the focal point, no real image is formed; only virtual rays appear to diverge from the focal point.
5. For objects closer than the focal point, virtual, upright images are formed on the same side as the object.

CONVEX MIRRORS

1. Convex mirrors always create virtual, upright images, regardless of object position.
2. These images are formed by diverging reflected rays, which appear to originate from a point behind the mirror (virtual focus).
3. The size of the image is smaller than the object.

IMAGE FORMATION BY CONCAVE MIRROR

OBJECT POSITION	IMAGE POSITION	IMAGE SIZE	NATURE OF IMAGE
Beyond C	Between C and F	Reduced	Real, Inverted
At C	At C	Same as Object	Real, Inverted
Between C and F	Beyond C	Enlarged	Real, Inverted
At F	No real image	No real image	No real image
Between F and Mirror	Virtual	Enlarged	Virtual, Upright



The spherical mirror equation is a mathematical relationship that relates the focal length (f), the object distance (u), and the image distance (v) for spherical mirrors. It helps determine the position and characteristics of the image formed by the mirror.

FORMULA

The spherical mirror equation can be expressed as:

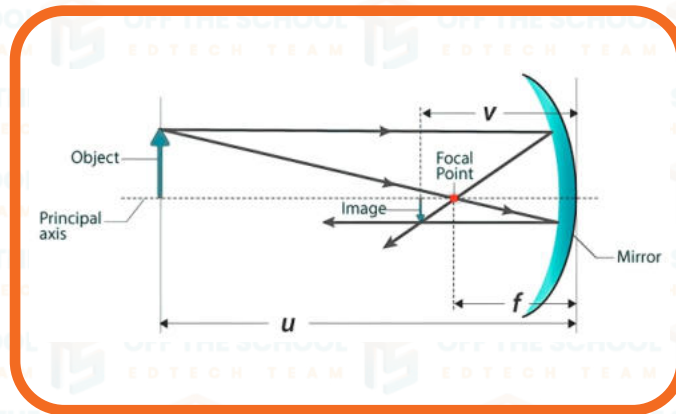
$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

Where:

(f) = focal length of the mirror

(u) = object distance (distance of the object from the mirror's pole)

(v) = image distance (distance of the image from the mirror's pole)



USES OF CONVEX MIRROR

Convex mirrors have several practical uses due to their unique reflective properties, such as the fact that they always produce virtual, upright, and diminished images. Some common applications include:

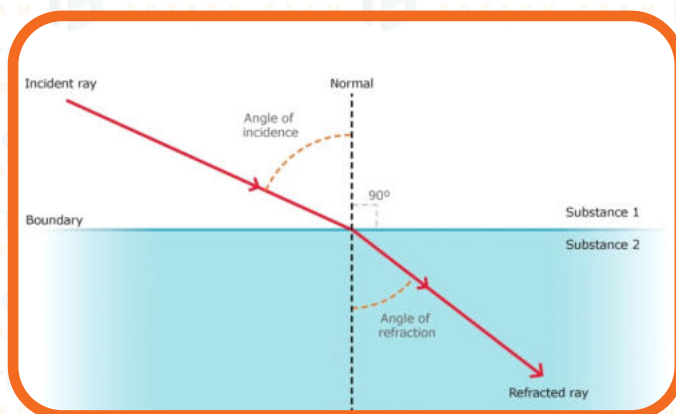
1. Convex mirrors are used as side-view and rearview mirrors in vehicles to provide a wider field of view, reducing blind spots and enhancing safety.
2. They are employed in security cameras and convex domes to monitor large areas, as they offer a broad panoramic view and minimize the risk of vandalism.
3. Convex mirrors are installed in stores to allow employees to observe multiple areas of the shop floor from a single vantage point, preventing theft and shoplifting.
4. Convex mirrors are used at sharp turns, intersections, and driveways to provide drivers with better visibility of oncoming traffic and pedestrians.
5. These mirrors are placed in parking lots to aid drivers in maneuvering, especially in areas with limited visibility.
6. Convex mirrors are used for decorative purposes, creating interesting visual effects and making spaces appear larger.
7. Some magnifying instruments, such as makeup mirrors, might use convex mirrors to provide a wider view while maintaining a smaller size.
8. Convex mirrors are often placed in outdoor settings, such as in gardens, to add aesthetics while also enhancing the feeling of spaciousness.

Convex mirrors play a crucial role in improving safety, visibility, and surveillance across various environments, making them valuable tools in everyday life.

REFRACTION OF LIGHT

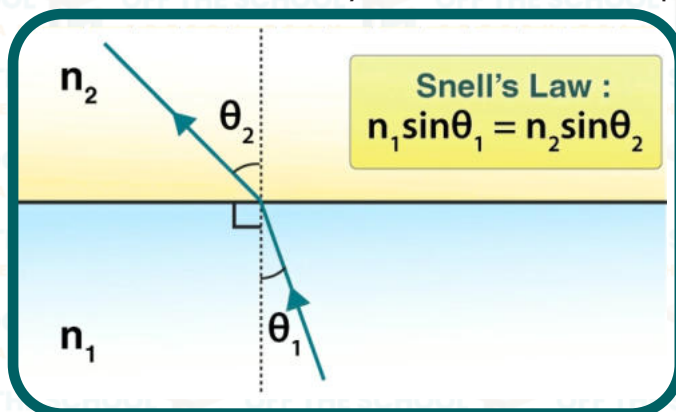
Refraction of Light refers to the bending of light rays as they pass from one transparent medium to another due to a change in their speed. This phenomenon occurs because light travels at different speeds in different mediums, causing a change in direction at the boundary between the two mediums.

Example: When a straw is partially submerged in a glass of water, the part of the straw submerged appears to be bent at the water's surface. This bending is due to the refraction of light as it travels from air to water, causing the light rays to change direction at the air-water boundary.



SNELL'S LAW

It is a fundamental principle in optics that describes the relationship between the angles of incidence and refraction when light passes from one medium to another. It quantifies the bending of light as it crosses the boundary between two transparent mediums.



FORMULA

Snell's Law is mathematically represented as:

$$\frac{\sin \theta}{\sin \theta} = \frac{v_1}{v_2}$$

Where:

θ_1 = Angle of incidence

θ_2 = Angle of refraction

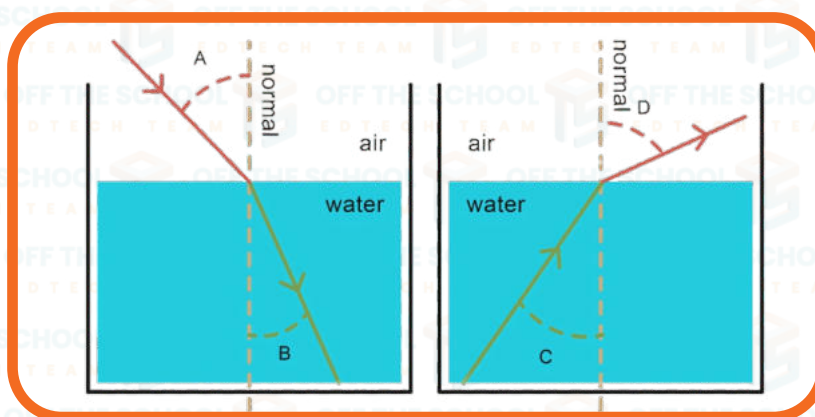
v_1 = Velocity of light in the first medium

v_2 = Velocity of light in the second medium

MATERIAL	REFRACTIVE INDEX	SPEED OF LIGHT (m/s)	CRITICAL ANGLE (degrees)
Water	1.333	2.25×10^8	~48.8
Glass (Crown)	1.5	2.00×10^8	~41.1
Diamond	2.42	1.24×10^8	~24.4
Acrylic	1.49	2.02×10^8	~42.1
Quartz	1.46	2.05×10^8	~46.6

TOTAL INTERNAL REFLECTION

Total internal reflection occurs when light traveling from a denser medium to a less dense medium is incident at an angle greater than the critical angle, causing all the light to be reflected back into the denser medium instead of refracting.



CRITICAL ANGLE

The critical angle is the angle of incidence at which light passing from a denser medium to a less dense medium experiences total internal reflection.

TELECOMMUNICATION THROUGH OPTICAL FIBER

Telecommunication via optical fiber involves transmitting data using light signals that travel through a thin, transparent fiber made of glass or plastic.

1. Information is converted into digital signals and sent as light pulses through the optical fiber.
2. Light is guided within the fiber through repeated total internal reflection, even when the fiber is curved or bent.
3. Optical fibers have low signal loss, ensuring data travels long distances without significant degradation.
4. Optical fibers provide high bandwidth, enabling fast data transmission rates for internet, phone calls, and multimedia.
5. Optical fibers are immune to electromagnetic interference, maintaining signal quality.
6. Multiple data streams can be transmitted simultaneously using different wavelengths of light (wavelength-division multiplexing).
7. Bundles of optical fibers are used to create cables for transmitting data over vast distances.

8. At destination points, light signals are converted back into digital data for interpretation.
9. Optical fibers offer secure communication as tapping the signals is difficult without disruption.
10. Optical fiber networks form the backbone of global communication, enabling the internet and worldwide communication networks.

REFRACTION THROUGH PRISM

Experiment: Dispersion of Light through a Prism

Materials:

- a. Prism (glass or acrylic)
- b. Light source (flashlight or laser pointer)
- c. White screen or wall

STEPS

1. Set up the experiment in a dark room with the light source and screen.
2. Place the prism on a surface in front of the light source.
3. Adjust the position of the prism so that a beam of light from the source falls on one face of the prism and emerges from the other face.
4. Observe the light passing through the prism and note the changes in its path.
5. Allow the refracted light to fall on the screen or wall.

OBSERVATIONS

1. As light enters the prism, it bends towards the base of the prism (towards the thicker part) due to refraction.
2. Inside the prism, the light undergoes dispersion, where different colors separate, forming a spectrum (rainbow-like pattern).
3. The order of colors in the spectrum is usually red, orange, yellow, green, blue, indigo, and violet (ROYGBIV).

CONCLUSION

The experiment demonstrates the dispersion of light as it passes through a prism. This dispersion occurs because different colors of light have different wavelengths, causing them to refract by varying degrees. The bending of light and the separation of colors illustrate how prisms and other optical devices can manipulate light and reveal its component colors.

This phenomenon is the basis for understanding rainbows, the behavior of light in various mediums, and the functioning of optical instruments.

IMAGE LOCATION BY LENS EQUATION

POWER LENS

Power of a Lens refers to the ability of a lens to converge or diverge light rays and is measured in diopters (D). It indicates the lens's focusing strength.

FORMULA

The formula to calculate the power of a lens is:

$$P = \frac{1}{f}$$

Where:

P is the power of the lens in diopters (D).

f is the focal length of the lens in meters (m).

For converging lenses (convex), the power is positive, indicating convergence. For diverging lenses (concave), the power is negative, representing divergence. The greater the power of a lens, the stronger its ability to bend light, either to focus or to spread it out.

SI Unit: The SI unit of lens power is the diopter (D), which is equivalent to one meter per focal length (1/m).

USES OF CONVEX LENSES

Magnifying Glass: A magnifying glass is a simple convex lens that is used to enlarge the apparent size of objects when viewed through it. It creates a virtual and magnified image of the object, making it easier to see fine details.

Camera: A camera is an optical device that captures and records images by allowing light to enter through a lens and onto a photosensitive surface, such as film or an image sensor. The captured light forms an image that can be stored or displayed electronically.

Projector: A projector is an optical device that projects enlarged images or videos onto a screen or surface. It uses a light source and a lens system to magnify and focus the image onto the projection surface, making it visible to a larger audience.

Photographic Enlarger: A photographic enlarger is a device used in darkrooms to create enlarged photographic prints from negatives. It projects light through a negative and a lens onto light-sensitive paper, producing a larger version of the original image.

RESOLVING POWER

Resolving power refers to the ability of an optical instrument, such as a microscope or telescope, to distinguish between two closely spaced objects as separate entities. It quantifies the instrument's ability to reveal fine details in an image.

MAGNIFYING POWER

Magnifying power is the factor by which an optical instrument, like a microscope or magnifying glass, enlarges the apparent size of an object when viewed through it.

FORMULA

The formula for magnifying power is given by:

$M = \text{size of the image} / \text{size of object}.$

$M = h_i / h_o$

MICROSCOPE

Microscopy: Microscopy is the science of using optical instruments to observe and study objects that are too small to be seen with the naked eye.

Simple Microscope: A simple microscope is a basic optical instrument that consists of a single convex lens used to magnify small objects.



The magnification provided by a simple microscope is calculated using the formula:

$$M = \frac{\theta_1}{\theta_2}$$

This formula quantifies how much larger the object appears when viewed through the simple microscope compared to when viewed with the naked eye.

COMPOUND MICROSCOPE

A compound microscope is an optical instrument that uses a combination of two lenses, an objective lens and an eyepiece lens, to magnify small objects. The objective lens produces a magnified real image that is further magnified by the eyepiece lens to provide a high total magnification.

MAGNIFICATION BY COMPOUND MICROSCOPE

The total magnification provided by a compound microscope is the product of the magnification of the objective lens and the magnification of the eyepiece lens.

FORMULA

$$M = \frac{L}{f_o} = \left(1 + \frac{25(cm)}{f} \right)$$

The magnification of the objective lens depends on its focal length and the tube length of the microscope, while the magnification of the eyepiece lens is often a fixed value provided by the manufacturer.

USES OF MICROSCOPE

1. Biological Research: Microscopes are extensively used in biological research to study the structure, function, and behavior of microscopic organisms, cells, tissues, and cellular components.

2. Medical Diagnosis: Microscopes play a crucial role in medical diagnosis by enabling the examination of blood cells, bacteria, parasites, and other microorganisms, aiding in disease identification and treatment.

3. Material Analysis: Microscopes are used to analyze materials at the micro and nanoscale, helping in fields like materials science, nanotechnology, and forensic science to understand material properties and structures.

TELESCOPE

A telescope is an optical instrument designed to observe distant objects by collecting and magnifying light. Telescopes come in various types, including refracting telescopes that use lenses and reflecting telescopes that use mirrors.

MAGNIFICATION BY TELESCOPE

The magnification provided by a telescope is calculated using the formula:

$$\text{Magnification} = \frac{\text{Focal length of Telescope's Objective lens/Mirror}}{\text{Focal Length of the Eyepiece Lens}}$$

This formula indicates how much larger the observed image appears compared to the naked eye, allowing astronomers to study celestial objects in greater detail.

USES OF TELESCOPE

1. Astronomical Observation: Telescopes are extensively used in astronomy to observe celestial objects such as stars, planets, galaxies, and nebulae, helping scientists study the universe's origin, composition, and evolution.

2. Space Exploration: Telescopes in space, like the Hubble Space Telescope, provide clear images without atmospheric distortion, contributing to our understanding of distant galaxies, exoplanets, and cosmic phenomena.

3. Scientific Research: Telescopes are employed in various scientific disciplines, from atmospheric studies to geodetic surveys, enabling researchers to monitor climate changes, track asteroids, and gather data for earth sciences.

THE HUMAN EYE AND DEFECT IN VISION

CORRECTION OF EYE DEFECTS BY LENS

Lenses are used to correct common eye defects such as myopia (nearsightedness) and hyperopia (farsightedness):

1. Myopia (Nearsightedness):

Correction: A concave (diverging) lens is used to diverge incoming light rays before they reach the eye's lens. This allows the focal point to move back onto the retina, enabling distant objects to be focused properly.

2. Hyperopia (Farsightedness):

Correction: A convex (converging) lens is used to converge incoming light rays before they enter the eye's lens. This shifts the focal point forward onto the retina, allowing close-up objects to be focused correctly.

By selecting the appropriate lens power, eye defects are corrected, ensuring that light focuses accurately on the retina for clear vision at various distances.